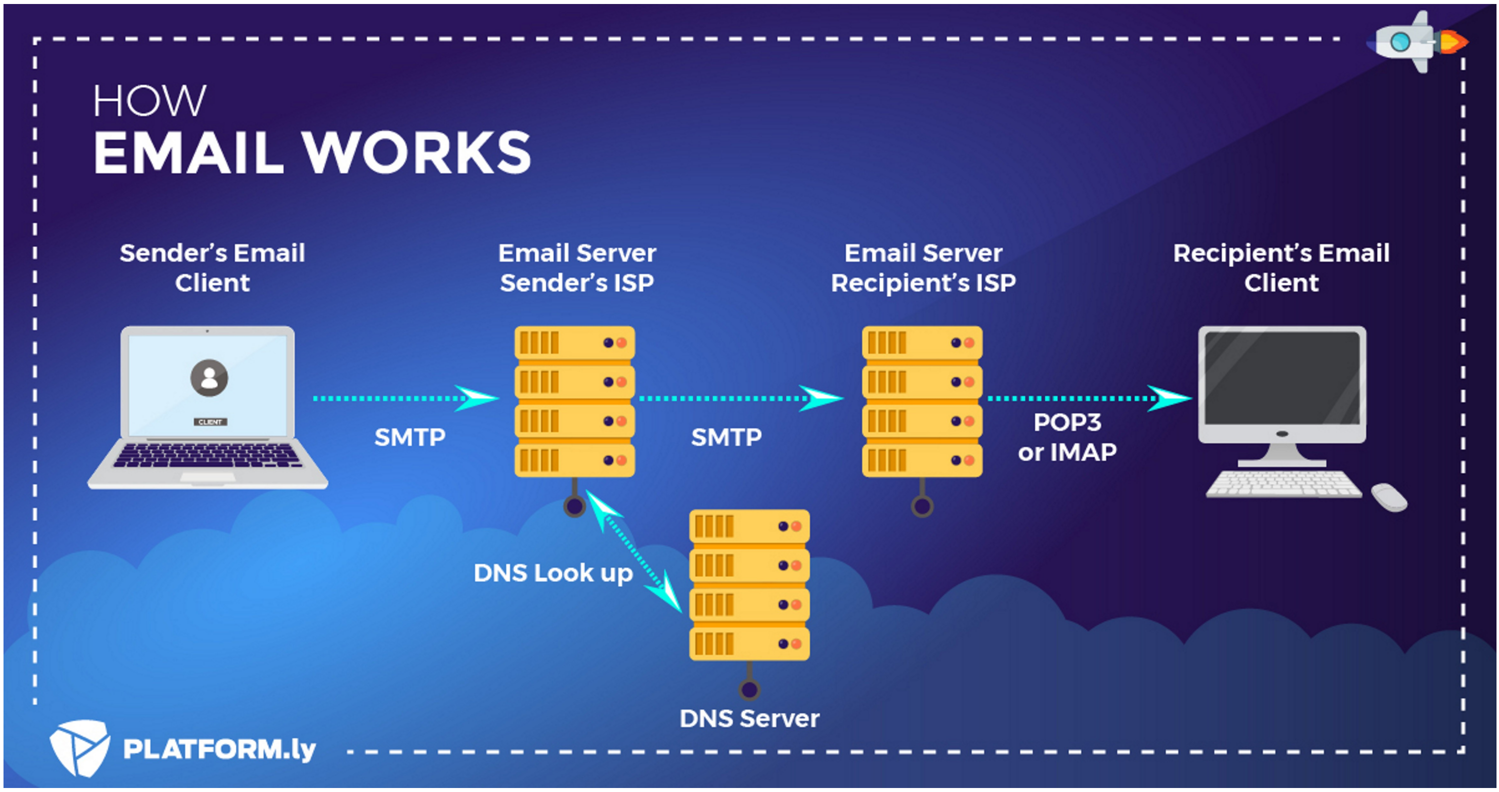
Tuesday, August 19, 2025 9:44 PM



📨 **What is an MX Record?**

MX (Mail Exchange) Record is a type of **DNS (Domain Name System) record** that tells the world **which mail server is responsible for receiving emails** for a particular domain.

👉 In short:

If DNS = phonebook of the internet, then MX record = "mail server address" for a domain.

🔹 **How it works**

1. You send an email to **bob@example.com**.
2. Your email server asks DNS:
   👉 "Hey, what mail server handles mail for example.com?"
3. DNS replies with the domain's **MX records** (one or more).
   Example:

```
example.com.    MX   10 mail1.example.com.
example.com.    MX   20 mail2.example.com.
```

4. The sending server then connects to the hostname(s) listed (mail1, mail2).

- mail1 has priority 10 → tried first.
- If it fails, it tries mail2 with priority 20.

🔹 **Structure of an MX Record**

- **Domain** — the domain that owns the record (example.com).
- **Priority** — a number (lower = higher priority).
- **Mail server hostname** — the server that accepts mail (must resolve to an A/AAAA record).

Example:

```
@    IN    MX    10   mail.example.com.
```

🔹 **Example in Real Life**

Check MX for Gmail:

```
$ nslookup -type=MX gmail.com
gmail.com   MX preference = 5, exchange = gmail-smtp-in.l.google.com
gmail.com   MX preference = 10, exchange = alt1.gmail-smtp-in.l.google.com
gmail.com   MX preference = 20, exchange = alt2.gmail-smtp-in.l.google.com
```

➡️ Gmail has multiple mail servers, with priorities 5, 10, 20, etc.

➡️ If the primary fails, others act as backup.

🔹 **Why MX Records are Important**

- Without MX, your domain cannot **receive email**.
- Multiple MX records = **redundancy** (backup servers).
- Email security checks (SPF, DKIM, DMARC) rely on proper MX setup.

✅ **In one line:**

MX record = DNS entry that directs where emails for your domain should be delivered.